

Lecture 6, Jan 05, 2025.

Rates of change. Suppose y is a quantity that depends on another quantity x , $y = f(x)$. If x changes from x_1 to x_2 , then the change in x is

$$\Delta x = x_2 - x_1$$

and the corresponding change in y is

$$\Delta y = f(x_2) - f(x_1).$$

the difference quotient

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

is called the average rate of change of y with respect to x over the interval $[x_1, x_2]$.

Def. Instantaneous rate of change $\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{x_2 \rightarrow x_1} \frac{f(x_2) - f(x_1)}{x_2 - x_1}$

Example: (For you). A manufacturer produces bolt of a fabric with a fixed width. The cost of producing x yards of this fabric is $C = f(x)$ dollars.

a). What is the meaning of the derivative $f'(x)$? What are its units?

b). In practical terms, what does it mean to say that $f'(1000) = 9$?

c). Which do you think is greater $f'(50)$ or $f'(500)$?

What about $f'(5000)$?

2.2. The Derivative as a function.

We know that

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Let $a = x$ then

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Given any number x for which this limit exists, we assign to x the number $f'(x)$, f' is a new function called the derivative of f

Example. a). if $f(x) = x^3 - x$ find a formula for $f'(x)$.

b). Illustrate this formula by comparing the graphs of f and f' .

Sln. a).

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f(x+h) = (x+h)^3 - (x+h) = x^3 + 3x^2h + 3xh^2 + h^3 - x - h$$

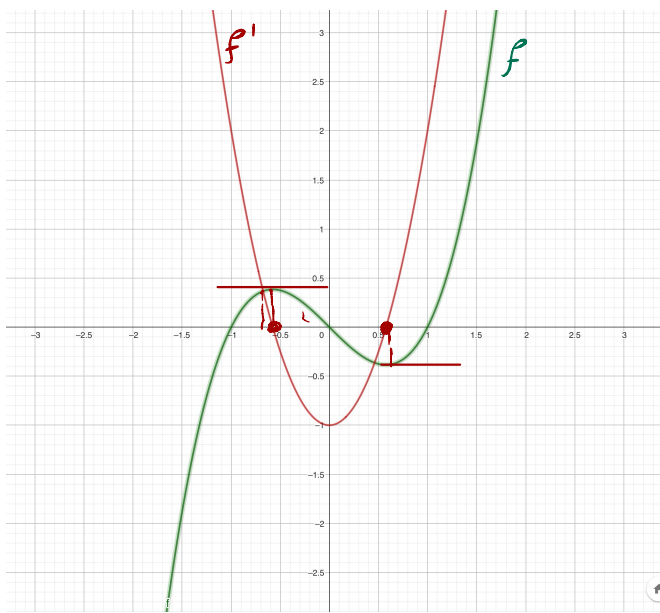
$$f(x) = x^3 - x$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\cancel{x^3} + 3x^2h + 3xh^2 + h^3 - \cancel{x} - h - \cancel{x^3} + \cancel{x}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3 - h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{h}(3x^2 + 3xh + h^2 - 1)}{\cancel{h}} = 3x^2 - 1$$

b).



Notice that $f'(a) = 0$ then f has horizontal tangents. and $f'(x) \geq 0$ when the tangents has positive slope. and $f'(x) \leq 0$ when the tangents has negative slopes.

Example. If $f(x) = \sqrt{x}$ find the derivative of f . State the domain of f' .

Sln.

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \times \frac{(\sqrt{x+h} + \sqrt{x})}{(\sqrt{x+h} + \sqrt{x})} \\
 &= \lim_{h \rightarrow 0} \frac{(\sqrt{x+h})^2 - (\sqrt{x})^2}{h(\sqrt{x+h} + \sqrt{x})} \\
 &= \lim_{h \rightarrow 0} \frac{x+h - x}{h(\sqrt{x+h} + \sqrt{x})} \\
 &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{1}{2\sqrt{x}}
 \end{aligned}$$

then $\text{Dom } f' = (0, \infty)$ and $\text{Dom } f = [0, \infty)$.

Problem find f' if $f(x) = \frac{1-x}{2+x}$

other Notations.

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}(f(x)) = Df(x) = D_x f(x).$$

the symbols $\odot$ and $\frac{dy}{dx}$ are called differentiation operators.

If we want to indicate the value of $\frac{dy}{dx}$ in the Leibniz notation at a particular number a .

$$\left. \frac{dy}{dx} \right|_{x=a} \quad \text{or} \quad \left. \frac{dy}{dx} \right]_{x=a} \quad \text{or} \quad f'(a).$$

Def. A function f is **differentiable at a** if $f'(a)$ exists. It is differentiable on an open interval (a, b) or (a, ∞) or $(-\infty, a)$ or $(-\infty, \infty)$ if it's differentiable at every number in the interval.

Example. where is the function $f(x) = |x|$ differentiable?

Sln. we know

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

if $x > 0$ then $|x| = x$ then

$$f'(x) = \lim_{h \rightarrow 0} \frac{|x+h| - |x|}{h} = \lim_{h \rightarrow 0} \frac{x+h - x}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1.$$

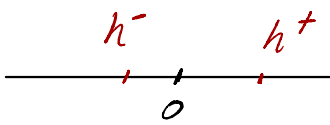
So, f is differentiable for every $x > 0$.

if $x < 0$: we have $|x| = -x$ so $x+h < 0$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{|x+h| - |x|}{h} = \lim_{h \rightarrow 0} \frac{-(x+h) - (-x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-x-h+x}{h} = \lim_{h \rightarrow 0} \frac{-h}{h} = -1. \end{aligned}$$

then f is differentiable for every $x < 0$

for $x=0$ then

$$f'(0) = \lim_{h \rightarrow 0} \frac{|0+h| - |0|}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h}$$


$$\lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1.$$

$$\lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1.$$

then $f'(0)$ does not exist. So f is not differentiable at $x=0$.

$$f'(x) = \begin{cases} -1 & \text{if } x < 0 \\ 1 & \text{if } x > 0. \end{cases}$$

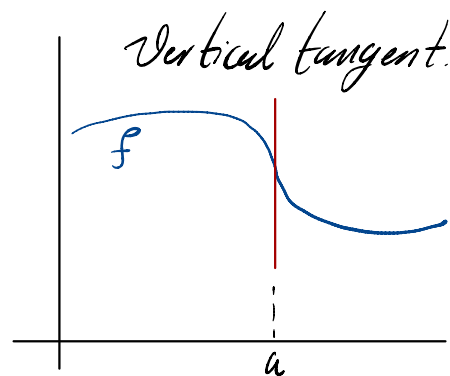
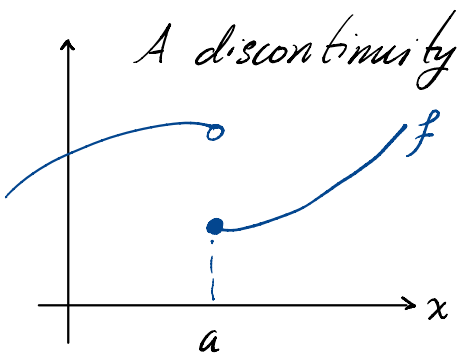
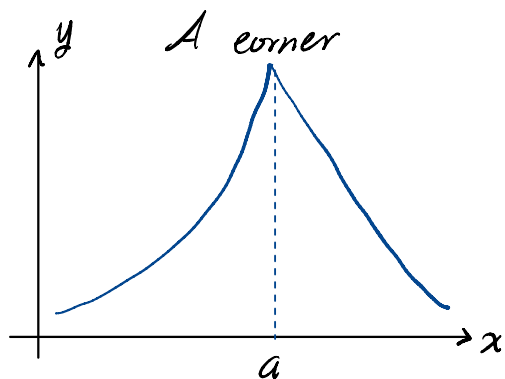
Thm. If f is differentiable at a then f is continuous at a .

The converse is not true.

⚠ if f is continuous at a then f is differentiable at a .

Example. $f(x) = |x|$ is continuous at $x=0$ but it's not differentiable at $x=0$.

Cases where $f'(a)$ does not exist.



Higher Derivatives.

If f is a differentiable function, then its derivative f' is also a function, so f' may have derivative of its own, denoted by $(f')' = f''$ this function is called the second derivative of f .

$$\underbrace{\frac{d}{dx}}_{\text{derivative of}} \left(\underbrace{\frac{dy}{dx}}_{\text{first derivative}} \right) = \underbrace{\frac{d^2 y}{dx^2}}_{\text{second derivative.}}$$

Example. if $f(x) = x^3 - x$ find $f''(x)$.

Sln. we know that $f'(x) = 3x^2 - 1$

$$f''(x) = (f')'(x) = \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(3(x+h)^2 - 1) - (3x^2 - 1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(3(x^2 + 2xh + h^2) - 1) - 3x^2 + 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{3x^2} + 6xh + 3h^2 - \cancel{1} - \cancel{3x^2} + \cancel{1}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{6xh + 3h^2}{h}$$

$$= \lim_{h \rightarrow 0} h \frac{(6x + 3h)}{h} = 6x$$

The third derivative f''' is the derivative of f''

$$y''' = f'''(x) = \frac{d}{dx} \left(\frac{d^2 y}{dx^2} \right) = \frac{d^3 y}{dx^3}$$

check that $f'''(x) = 6$.

2.3. Differentiation Formulas.

* Constant Functions. Let $f(x) = c$ where $c \in \mathbb{R}$ is in general a constant function.

Derivative of a constant function

$$\frac{d}{dx}(c) = \frac{d}{dx}(f(x)) = f'(x) = 0$$

Pf. $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0.$

* Power functions. Let $f(x) = x^n$ for $n \in \mathbb{N}$.

The Power rule: if n is a natural number.

$$\frac{d}{dx}(x^n) = \frac{d}{dx}(f(x)) = f'(x) = n x^{n-1}.$$

PF. we know that

$$(x^n - a^n) = (x-a)(x^{n-1} + x^{n-2}a + \dots + xa^{n-2} + a^{n-1}).$$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a}$$

$$= \lim_{x \rightarrow a} \frac{(x-a)(x^{n-1} + x^{n-2}a + \dots + xa^{n-2} + a^{n-1})}{(x-a)}$$

$$= a^{n-1} + a^{n-1} + \dots + a^{n-1}$$

$$= n a^{n-1}$$

$$\text{So, } f'(x) = n x^{n-1}.$$

Example.

- $f(x) = x^6$ $f'(x) = 6 x^{6-1} = 6 x^5$
- $f(x) = x^{1000}$ $f'(x) = 1000 x^{999}$.

New Derivatives from old.

The constant Multiple rule: If c is a constant and f is differentiable

$$\text{then } \frac{d}{dx}[c f(x)] = c \cdot \frac{d}{dx}(f(x)).$$

Example. Compute f' where $f(x) = \frac{13}{7} x^{15}$.

Sln:

$$f'(x) = \frac{d}{dx} \left(\underbrace{\frac{13}{7}}_c x^{15} \right) = \frac{13}{7} \underbrace{\frac{d}{dx}(x^{15})}_{\text{power rule}} = \frac{13}{7} \cdot 15 x^{14} = \frac{195}{7} x^{14}.$$

Sum of functions. Let f and g be differentiable functions.

$$\frac{d}{dx}(f(x) \pm g(x)) = \frac{d}{dx}(f(x)) \pm \frac{d}{dx}(g(x)) = f'(x) \pm g'(x)$$

Example. Compute $\frac{d}{dx} (x^8 + 12x^5 - 4x^4 + 10x^3 - 6x + 5)$.

Sln. $\frac{d}{dx} (x^8 + 12x^5 - 4x^4 + 10x^3 - 6x + 5) = \frac{d}{dx} (x^8) + \frac{d}{dx} (12x^5) - \frac{d}{dx} (4x^4)$
 $+ \frac{d}{dx} (10x^3) - \frac{d}{dx} (6x) + \underbrace{\frac{d}{dx} (5)}_0$
 $= 8x^7 + 12 \frac{d}{dx} (x^5) - 4 \frac{d}{dx} (x^4) + 10 \frac{d}{dx} (x^3) - 6 \underbrace{\frac{d}{dx} (x)}_1$
 $= 8x^7 + 12(5x^4) - 4(4x^3) + 10(3x^2) - 6$
 $= 8x^7 + 60x^4 - 16x^3 + 30x^2 - 6.$

Example. Find the points on the curve $y = x^4 - 6x^2 + 4$ where the tangent line is horizontal.

Sln. Horizontal tangents occur where $y' = 0$.

$$y' = \frac{dy}{dx} = \frac{d}{dx} (x^4) - 6 \frac{d}{dx} (x^2) + \frac{d}{dx} (4)$$
$$= 4x^3 - 12x$$

then $4x^3 - 12x = 0$

$$4x(x^2 - 3) = 0$$

then at $x=0, x=\sqrt{3}, x=-\sqrt{3}$

if $x=0$ $f(0) = 4 \Rightarrow (0, 4)$

if $x=\sqrt{3}$ $f(\sqrt{3}) = -5 \Rightarrow (\sqrt{3}, -5)$

if $x=-\sqrt{3}$ $f(-\sqrt{3}) = -5 \Rightarrow (-\sqrt{3}, -5)$

} here we have horizontal tangents.

The product Rule. if f and g are differentiable, then

$$\frac{d}{dx} (f \cdot g) = f'(x) \cdot g(x) + f(x) \cdot g'(x).$$

Rmk. $(f \cdot g)' \neq f' \cdot g' \quad \nabla$

Example. Find $F'(x)$ if $F(x) = (6x^3)(7x^4)$

Sln. Notice that $F(x) = f(x) \cdot g(x)$ where $f(x) = 6x^3$, $g(x) = 7x^4$.

Using the product rule.

$$\begin{aligned} F'(x) &= f'(x) \cdot g(x) + f(x) \cdot g'(x) & \begin{cases} f'(x) = 18x^2 \\ g'(x) = 28x^3 \end{cases} \\ &= 18x^2(7x^4) + (6x^3)(28x^3) \\ &= 168x^6 + 168x^6 = 336x^6 \end{aligned}$$

Example. If $h(x) = x \cdot g(x)$ and it is known $g(3) = 5$, $g'(3) = 2$ then compute $h'(3)$.

Sln. Using the product rule.

$$h'(x) = \frac{d}{dx}(x) \cdot g(x) + x \cdot g'(x)$$

$$h'(x) = g(x) + x g'(x)$$

then

$$h'(3) = g(3) + 3g'(3)$$

$$h'(3) = 5 + 3(2) = 11$$

The Quotient Rule. If f and g are differentiable functions

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{(g(x))^2}$$

Example. Compute y' where $y = \frac{x^2 + x - 2}{x^3 + 6}$

Sln. $y = \frac{f(x)}{g(x)}$; $f(x) = x^2 + x - 2$, $g(x) = x^3 + 6$
 $f'(x) = 2x + 1$, $g'(x) = 3x^2$.

by the Quotient rule

$$\begin{aligned}
 y' &= \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} = \frac{(2x+1)(x^3+6) - (x^2+x-2)(3x^2)}{(x^3+6)^2} \\
 &= \frac{2x^4 + 12x + x^3 + 6 - 3x^4 - 3x^3 + 6x^2}{(x^3+6)^2} \\
 &= \frac{-x^4 - 2x^3 + 6x^2 + 12x + 6}{(x^3+6)^2}.
 \end{aligned}$$

Rmk. Don't use the quotient rule every time you see a quotient.
for example let

$$F(x) = \frac{3x^2 + 2\sqrt{x}}{x}$$

Using algebra we know $\left(\frac{x+y}{a} = \frac{x}{a} + \frac{y}{a} \right)$ then,

$$F(x) = \frac{3x^2}{x} + \frac{2\sqrt{x}}{x} = 3x + 2x^{-1/2}.$$

Rmk. if $n \in \mathbb{R}$ then

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Example. if $f(x) = x^\pi$ then $f'(x) = \pi x^{\pi-1}$.

• Let $y = \frac{1}{\sqrt[3]{x^2}}$ we know $\sqrt[n]{x} = x^{1/n}$ $x^{-n} = \frac{1}{x^n}$

$$y = \frac{1}{x^{2/3}} = x^{-2/3}$$

$$y' = -\frac{2}{3} x^{-2/3-1} = -\frac{2}{3} x^{-5/3} = -\frac{2}{3x^{5/3}}.$$

Example. Differentiate the function $f(t) = \sqrt{t}(a+bt)$

Sln. using the product rule $g(t) = \sqrt{t}$ $h(t) = a+bt$
 $g'(t) = \frac{1}{2\sqrt{t}}$ $h'(t) = b.$

$$f'(t) = g'(t)h(t) + g(t)h'(t) = \frac{1}{2\sqrt{t}} \cdot (a+bt) + \sqrt{t}(b)$$

$$= \frac{a+bt}{2\sqrt{t}} + \frac{b\sqrt{t}}{1} = \frac{a+bt + 2b(\sqrt{t})^2}{2\sqrt{t}} = \frac{a+bt + 2bt}{2\sqrt{t}}$$

$$= \frac{a+3bt}{2\sqrt{t}}.$$